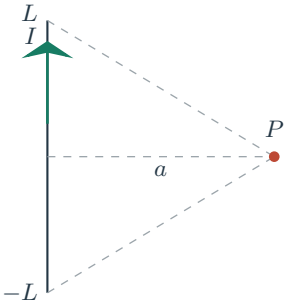
M5-023 有限直导线及其极限



$d\mathbf{l} \times \mathbf{R}$ 在 P 指向纸内

$$B = \frac{\mu_0 I L}{2\pi a\sqrt{L^2 + a^2}}$$

$$L \ll a : \quad B \simeq \frac{\mu_0 I L}{2\pi a^2}$$

$$L \gg a : \quad B \to \frac{\mu_0 I}{2\pi a}$$